

Each discrete tiny packet of energy is quantum. And the bundle of light energy is called photon.

Planck quantum formula

In 1900, Max Planck explained the particle nature of light with his theory known as Planck's quantum theory.

$$(E = h\nu)$$

For a single photon

$$(E = nh\nu)$$

For "n" no of photons

Units of Planck quantum

$$E = \frac{hc}{\lambda}$$

(Or)

$$E = \frac{(6.626 \times 10^{-34}) \times (3 \times 10^8)}{\lambda}$$

$$E = \frac{19.878 \times 10^{-26}}{\lambda}$$

Planck wavelength equation

$$E = hc\tilde{\nu}$$

(Or)

$$E = (6.626 \times 10^{-34}) \times (3 \times 10^8) \times \tilde{\nu}$$

$$E = (19.878 \times 10^{-26}) \times \tilde{\nu}$$

Planck wavenumber equation

SI Unit--- Joule

CGS unit--- Erg

Atomic unit ---
electron volt (eV)

Jayam chemistry adda

Blackbody Radiation Problem #1

- Power output by a radiating body as a function of temperature is given by the function to the right.

- Calculate the Sun's temperature assuming it can be modeled as a perfect blackbody.
- Calculate its wavelength of maximum emission using Wien's displacement law.

$$T = \sqrt[4]{\frac{P_{\text{sun}}}{4\pi r_{\text{sun}}^2 \sigma e}} = \sqrt[4]{\frac{(3.85 \times 10^{26} \text{ W})}{4\pi (6.96 \times 10^8 \text{ m})^2 (5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}) (1)}}$$

$$T = 5779 \text{ K}$$

$$\lambda_{\text{max}} T = 0.2898 \times 10^{-2} \text{ m} \cdot \text{K}$$

$$\lambda_{\text{max}} = \frac{0.2898 \times 10^{-2} \text{ m} \cdot \text{K}}{5779 \text{ K}}$$

$$\lambda_{\text{max}} = 5.01 \times 10^{-7} \text{ m} = 501 \text{ nm}$$

$$P = A\sigma eT^4$$

$$\sigma = 5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}$$

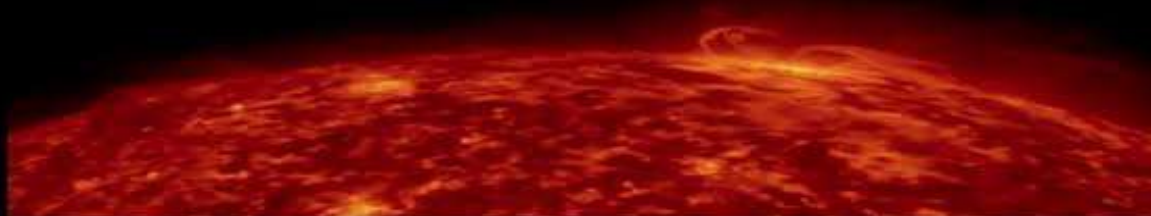
$$0 \leq e \leq 1$$

$$P_{\text{sun}} = 3.85 \times 10^{26} \text{ W}$$

$$r_{\text{sun}} = 6.96 \times 10^8 \text{ m}$$

$$A_{\text{sphere}} = 4\pi r^2$$

- According to **Wien's displacement law**, the product of wavelength of maximum emission and the temperature at which that occurs is a constant.



Problem-1: Calculate the energy of a photon at 525 nm.

Solution:

The Wavelength of photon=525 nm

The formula to measure the energy of photon is;

$$E = \frac{19.878 \times 10^{-26}}{\lambda}$$

$$E = \frac{19.878 \times 10^{-26}}{525 \times 10^{-9}} \text{ joule}$$

$$E = 0.0378 \times 10^{-17} \text{ joules}$$

$$E = 3.78 \times 10^{-19} \text{ joules}$$

3

Jayam chemistry adda



Problem-2: The energy absorbed by each molecule (A_2) of a compound is 8.4×10^{-19} joules. And bond energy per molecule is 8.0×10^{-19} joules. What is the kinetic energy per atom of the molecule?

Solution:

Energy absorbed by a molecule = 8.4×10^{-19} joules

Bond energy of each molecule of the substance = 8.0×10^{-19} joules

The formula to calculate the kinetic energy per atom of the molecule is;

$$\text{Kinetic energy of each atom of a molecule} = \frac{(\text{Energy absorbed}) - (\text{bond energy per molecule})}{\text{number of atoms in the molecule}}$$

$$KE = \frac{(8.4 \times 10^{-19}) - (8.0 \times 10^{-19}) \text{ joules}}{2}$$

$$KE = 0.2 \times 10^{-19} \text{ joules} = 2 \times 10^{-20} \text{ joules}$$

4

Jayam chemistry adda



Problem-3: Find the number of photons of light at 4000 pm having 1 joule of energy ?

Solution:

Energy of light = 1 joule

Wavelength of light = 4000 pm = 4000×10^{-12} m

The formula to measure the number of photons of light is;

$$E = nh\nu = \frac{nhc}{\lambda}$$

$$n = \frac{\lambda E}{hc} = \frac{\lambda E}{(19.878 \times 10^{-26}) \text{ Jm}} \quad (\text{By substituting the values of h and c})$$

$$n = \frac{(4000 \times 10^{-12} \text{ m}) \times 1 \text{ J}}{(19.878 \times 10^{-26}) \text{ Jm}}$$

$$n = 201.22 \times 10^{14} = 2.0122 \times 10^{16}$$

5

Jayam chemistry adda



Problem-4: A 242 nm electromagnetic light can ionize a sodium atom. What is the ionization energy of sodium in KJ/mole?

Solution:

The wavelength of electromagnetic light = 242 nm = 242×10^{-9} m

The formula to calculate the ionization energy of one sodium atom is;

$$E = \frac{(19.878 \times 10^{-26}) \text{ Jm}}{\lambda}$$

$$E = \frac{(19.878 \times 10^{-26}) \text{ Jm}}{242 \times 10^{-9} \text{ m}}$$

$$E = 0.0821 \times 10^{-17} \text{ Joule}$$

Ionization energy of one mole of a substance = ionization energy of one atom $\times$ Avogadro's number

Ionization energy of one mole of sodium = $(0.0821 \times 10^{-17} \text{ joule}) \times (6.023 \times 10^{23})$

Ionization energy of one mole of sodium = $0.494 \times 10^6 \text{ joule/mole} = 494 \text{ KJ/mole}$

6

Jayam chemistry adda

